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Text2Screen: Words to Visuals in Motion

- **Problem Statement ID – 1600**
- **Problem Statement Title- Student Innovation**
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team ID-**
- **Team Name (Registered on portal)- VidGen Innovators**

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Text2Screen: Words to Visuals in Motion

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❖Proposed Solution(Describe your Idea/Solution/Prototype)

Proposed Solution:

* **Automated Content Transformation:** Converts text into video automatically.
* **NLP and Multimedia Processing:** Ensures context-aware and visually appealing presentations.
* **User-Friendly Interface:** Easy for non-technical users.

Addressing the Problem:

* **Scalability:** Automates video generation, saving time and resources.
* **Increased Engagement:** Videos are more engaging and accessible than text.
* **Flexibility:** Works with various text sources like PIB releases, blog posts, and news articles.

Innovation and Uniqueness:

* **Dynamic Content Summarization:** Use AI to summarize long texts for concise video output.
* **Adaptive Video Lengths:** Adjust video length for different platforms automatically.
* **Language & Localization:** Support Indian language videos with localized visuals and narration.
* **Dynamic video duration features**: Allow users to specify desired video duration.

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TECHNICAL APPROACH

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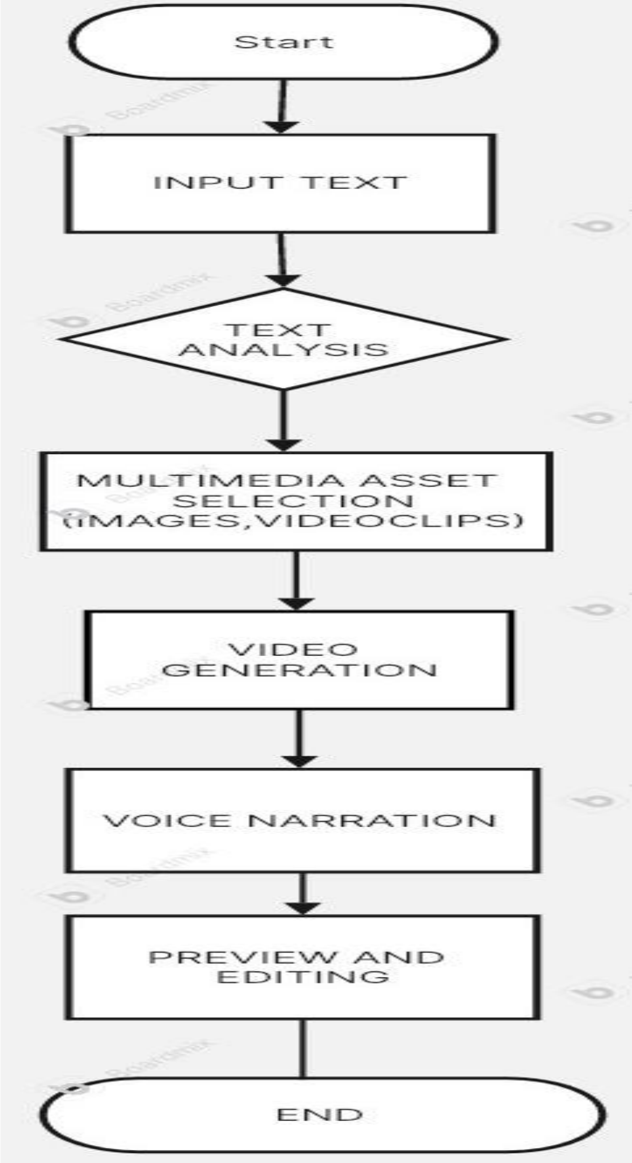
- **Backend:** Python
  Handles server-side logic and integrates various components.
- **Web Framework:** Streamlit
  Provides an interactive interface for user engagement.
- **Web Scraping:** Beautiful Soup or Scrapy
  Web scraping libraries for extracting text content from websites.
- **NLP:** NLTK or SpaCy
  Performs text analysis, including tokenization and entity recognition.
- **Video Editing:** MoviePy
  Generates and edits videos programmatically from text and multimedia.
- **Image Processing:** Pillow or OpenCV
  Manipulates and processes images and graphics.
- **Cloud Storage:** AWS S3 or Google Cloud Storage
  Stores and manages multimedia assets and generated videos.
- **Deployment:** Heroku or AWS Lambda
  Hosts the application and backend services, ensuring scalability.



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FEASIBILITY AND VIABILITY

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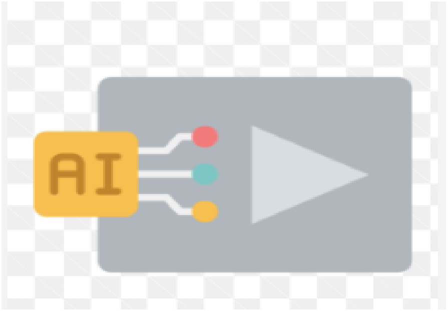
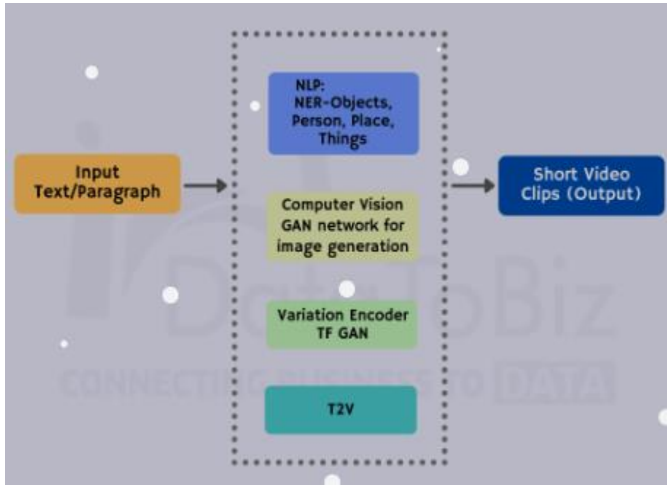
Technical Feasibility:



- **Existing Models:** AI models like DALL-E, and Imagen generate images from text, while Make-A-Video and Phenaki focus on videos but need high computational resources and are limited in video length and complexity.
- **Data Requirements:** Requires large, diverse datasets with paired text-video data, which are challenging to acquire and curate.
- **Computational Resources:** Demands high-performance GPUs, significant storage, and scalable cloud infrastructure.

Challenges and Risks:



- ❖ **Technical:** Ensuring high-quality, coherent videos, and allowing for customization is difficult. Generating longer videos is resource-intensive.
- ❖ **Ethical/Legal:** Risks include copyright issues, misinformation, and biases in generated content.

Strategies:

- ✓ Combine pre-trained models for better coherence and quality.
- ✓ Enable users to adjust video length, tone, and style.
- ✓ Secure content licenses or restrict to non-copyrighted/user-provided assets.
- ✓ Allow users to customize video components like backgrounds and animations.

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IMPACT AND BENEFITS

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Potential Impact on the Target Audience:

- **Content Creators:** Simplifies video creation, allowing them to focus more on creativity and less on technical aspects.
- **Educators:** Enhances teaching by easily generating interactive and visually engaging learning materials.
- **Marketers:** Boosts campaign effectiveness through automated, personalized video content tailored to specific audiences.
- **News Outlets:** Speeds up news delivery by automating the creation of visually rich content, improving audience reach.

Benefits of the Solution:

- **Social:** Broadens access to video production, enabling diverse voices to create and share content without needing specialized skills.
- **Economic:** Lowers costs and accelerates content production, driving efficiency for businesses and individuals.
- **Environmental:** Promotes sustainability by reducing the reliance on physical production processes, leveraging digital solutions instead.

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RESEARCH AND REFERENCES

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- A. Singh, "A Survey of AI Text-to-Image and AI Text-to-Video Generators," 2023 4th International Conference on Artificial Intelligence, Robotics and Control (AIRC), Cairo, Egypt, 2023, pp. 32-36, doi: 10.1109/AIRC57904.2023.10303174
- J. Jin, J. Wu, Z. Xu, H. Zhang, Y. Wang and J. Yang, "Text To Video: Enhancing Video Generation Using Diffusion Models And Reconstruction Network," 2023 2nd International Conference on Computing, Communication, Perception and Quantum Technology (CCPQT), Xiamen, China, 2023, pp. 108-114, doi: 10.1109/CCPQT60491.2023.00024
- D. Kim, D. Joo and J. Kim, "TiVGAN: Text to Image to Video Generation With Step-by-Step Evolutionary Generator," in IEEE Access, vol. 8, pp. 153113-153122, 2020, doi: 10.1109/ACCESS.2020.3017881.
- S. Raja, M. S and P. J, "Text to Video Generation using Deep Learning," *2023 Eighth International Conference on Science Technology Engineering and Mathematics (ICONSTEM)*, Chennai, India, 2023,pp.1-7,doi:10.1109/ICONSTEM56934.2023.10142725.

- https://paperswithcode.com/paper/videocomposer-compositional-video-synthesis
- https://paperswithcode.com/paper/modelscope-text-to-video-technical-report

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TEAM DETAILS

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